

CYBERHOST: A ONE-STAGE DIFFUSION FRAMEWORK FOR AUDIO-DRIVEN TALKING BODY GENERATION

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Background: Generative Human Video is Transforming the World

- We see avatars and virtual presenters reshaping education, media, and customer service.
- We leverage diffusion-based video generation to enable photorealistic, controllable human synthesis.
- We use audio-driven control as a natural interface that promises scalable content creation.

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The Problem: Audio-Driven Talking Body Generation is Underexplored

- We aim to generate half-body videos from a single image + speech with natural motion.
- We must preserve hand integrity, identity consistency, and stable body motion.
- We observe that existing methods blur hands, drift identity, or produce unstable gestures.



Figure: 现有V2V/A2V人体动画难以还原手部与面部细节; 本方法保持手部完整与面部身份一致性

Why Existing Pipelines Struggle

- We find two-stage systems rely on pose/mesh, adding complexity and cascading errors.
- We observe audio correlates weakly with full-body motion → higher uncertainty than talking heads.
- We note local regions (hands/face) dominate perception yet are underfit by global models.

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- We propose CyberHost, a one-stage diffusion framework for audio-driven talking bodies.
- We introduce RAM to inject local priors and identity descriptors for hands/face.
- We add human-prior-guided conditions to stabilize motion and reduce uncertainty.
- We show zero-shot video from a single image + audio, extensible to video/hybrid controls.

Region Attention Module (RAM): What We Add and Why

- We fuse a learnable spatio-temporal latent bank (local priors) with identity descriptors from cropped face/hands.
- We predict regional masks to focus updates and add auxiliary local losses (e.g., hand keypoint heatmaps).
- We find RAM improves hand integrity and facial identity without hurting global coherence.

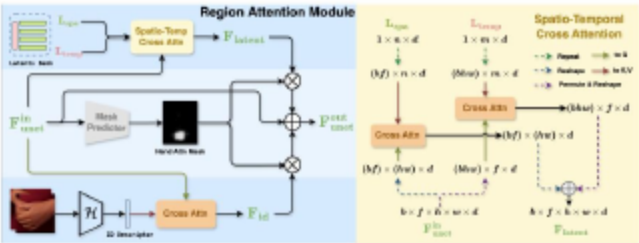


Figure: 区域注意力模块 (RAM) 示意: 手部区域

- We use a latent diffusion model with a 2D U-Net augmented by temporal modules.
- Inputs: one reference image, multi-scale Wav2Vec audio; optional short motion frames for continuation.
- We condition via cross-attention and a reference branch; RAM refines local regions; human priors stabilize global/local motion.
- Training: we pretrain on image consistency and then fine-tune end-to-end on audio-driven clips.

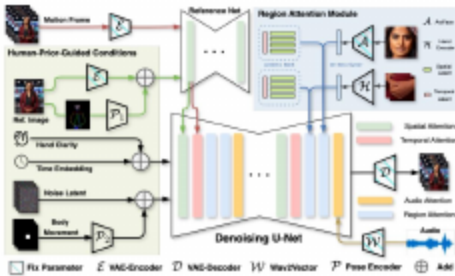


Figure: CyberHost: 音频驱动参考图像生成视频 U-Net 引入区域注意力 人体先验

Human Priors: How We Stabilize Motion

- We regularize global movement with a body-movement map.
- We inject a hand-clarity score to combat blur and control sharpness.
- We fuse pose-aligned reference features to inject skeletal topology.



Figure: Medium Clarity Score

- Tasks: A2V-B (audio→body), V2V-B (video→body), A2V-H (talking head).
- Baselines: DiffTED, DiffGesture+MimicMotion; DisCo, AnimateAnyone, MimicMotion; EchoMimic, VExpress, Hallo.
- Metrics: FID/FVD, CSIM, SyncC/SyncD, HKC/HKV, AKD.
- Evaluation includes standard splits (e.g., Vlogger test set) where applicable.

- We substantially reduce FID/FVD and improve CSIM and lip-sync over prior audio-driven methods.
- We improve hand integrity and motion richness (higher HKC/HKV).
- On the Vlogger test set, we also see stronger identity and sync.
- User study: we are preferred in identity, motion naturalness, and lip-sync.

Qualitative: Audio-Driven Talking Body

- We show sharper faces and clearer hands than baselines.
- We observe more natural, speech-synchronized gestures and better identity stability.



Figure: The audio-driven talking body results of CyberHost compared to other methods.

Video-Driven Body Reenactment (V2V-B): Highlights

- We outperform DisCo, AnimateAnyone, and MimicMotion in both image and temporal quality.
- We achieve lower pose error (AKD), indicating stronger structure tracking.
- We generalize beyond audio-only driving with consistent identity and fewer artifacts.

- We maintain structure and identity under diverse poses with fewer artifacts.
- We recover fine local details while preserving global motion stability.



Figure: Comparisons with other video-driven body reenactment results

- We combine explicit hand keypoints with audio for finer control.
- We generalize zero-shot to diverse identities and styles.
- We handle complex hand interactions with natural expressiveness.



Figure: 开放集测试图像: (a)多模态驱动(b)音频驱动

Ablation Insight: Where to Place RAMs

- We find inserting RAMs across all layers performs best.
- We observe that even layers with less-accurate masks contribute complementary refinements.
- We attribute the gains to distributed local updates that reinforce coherence at multiple scales.

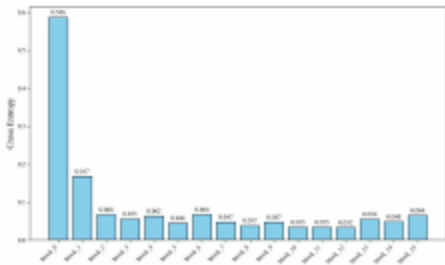


Figure: Hand-attention mask accuracy across denoising U-Net layers

Limitations and Failure Cases

- We can fail on stylized/cartoonish bodies or exaggerated proportions where priors break.
- We can be confused by skin-toned backgrounds that interfere with hand segmentation.
- We degrade under extreme poses or noisy/atypical audio (e.g., shouting/singing).



Figure: Failure cases of CyberHost in challenging scenarios.

- We present a one-stage audio-driven diffusion model that improves hands, identity, and motion stability.
- We achieve state of the art across audio- and video-driven tasks with zero-shot generation.
- Next: we pursue higher-resolution full-body synthesis, stronger robustness, and responsible deployment (watermarking/moderation).

- We thank our collaborators, data contributors, and the community baselines.
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